

Mathematics

IMPORTANT:

This resource was developed to support a previous version of the syllabus. It may contain content that differs from the current syllabus.

HSC Marking Feedback 2019

Question 11

Students should:

- use a calculator to solve trigonometric equations (a)
- round off to the given number of decimal places (a)
- use the Reference Sheet to obtain correct rules for derivatives (b) and (c)
- have a sound understanding of how to use the limiting sum formula (d)
- sketch a parabola and a line showing the given points of intersection (g)
- understand the difference between a linear and quadratic inequality (g)
- shade the region simultaneously satisfying two inequalities (g).

In better responses, students were able to:

- use the Reference Sheet and substitute correctly (a)
- use the appropriate mode function on the calculator (a)
- correctly use brackets and order of operations in algebraic expressions (c)
- understand the implications of a positive and negative ratio (d)
- use the correct order when substituting limits (e)
- set up a probability tree diagram (f)
- understand the difference between events with and without replacement (f)
- sketch a diagram using at least $\frac{1}{3}$ of a page (g).

Areas for students to improve include:

- having a sound understanding of the use of the sine rule (a)
- differentiating trigonometric expressions using the product rule (b)
- differentiating functions using the quotient rule (c)
- rearranging a fraction to obtain a primitive involving a negative power (e)
- remembering to include the limits of integration once the primitive is found (e)
- finding the probability of multiple events in questions that involve situations without replacement (f).

Question 12

Students should:

- use $y - y_1 = m(x - x_1)$ to find the equation of a line (ai)
- find x -intercepts by substituting $y = 0$ into the equation of the line (aii)
- be able to simply find the distance between two points on the x -axis (aii)
- obtain equations in a and d correctly using the formulae for T_n and S_n for an arithmetic series (b)
- competently substitute and use a calculator to evaluate an exponential expression (ci) and (cii)
- use logs to solve exponential equations (ciii)
- use definite integrals to find areas (d).

In better responses, students were able to:

- identify the gradient from the general form of a line (ai)
- correctly use $m_1 \times m_2 = -1$ to find gradient of a perpendicular line (ai)
- recognise that the y -coordinate of a point gives the height above the x -axis so there is no need to use perpendicular distance formula (aii)
- recognise that the distance between two points along the x -axis does not require the distance formula (aii)
- use the Reference Sheet to obtain the correct rules for arithmetic series (b)
- solve $9d = 3$ to obtain $d = \frac{1}{3}$ and not $d = 3$ (b)
- understand the link between the first derivative and rate of change (cii)
- differentiate exponentials with negative decimal powers and large coefficients (cii)
- achieve accuracy of calculations particularly when small decimals or fractions are involved (ciii)
- find the correct fraction in front of the integral in order to create a numerator which is the derivative of the denominator (d).

Areas for students to improve include:

- understanding the need for perpendicular height to find the area of triangle (aii)
- recognising that the y -coordinate of a point gives the height above the x -axis (aii)
- calculating the area of a triangle using different methods (aii)
- looking for simple methods instead of long algebraic process (aii)
- solving simultaneous linear equations competently (b)
- showing the substitution steps and rounding carefully (ci and cii)
- understanding that the rate of change is found using the first derivative (cii)
- substituting competently and evaluating exponential expressions accurately (cii)
- integrating correctly to reach a logarithmic function (d)
- substituting limits correctly (d)

- showing all working (d)
- reading the question carefully (d)
- using brackets accurately (d).

Question 13

Students should:

- use the correct mode when solving trigonometric equations (a)
- identify the correct quadrants from the sign of the trigonometric equation (a)
- use the arc length formula after converting degrees to radians (b)
- find the perimeter of a segment by calculating the length of all sides (b)
- find the volume of a solid of revolution using the correct method (d)
- square a cubic equation to find the primitive (d)
- graph an absolute value function given a specific domain (ei)
- use a graphical representation to solve an algebraic equation (eii).

In better responses, students were able to:

- rearrange a trigonometric equation by equating to zero (a)
- understand that $\sin x = 0$ has solutions in the given domain and consequently it is inappropriate to divide by this function (a)
- use the sine graph to solve for boundary angles (a)
- use a related formula to find the arc length and chord length (b)
- use the correct mode on the calculator (b)
- show the appropriate substitution into the arc length or cosine formulae (b)
- distinguish between the perimeter and the area of the shaded minor segment (b)
- demonstrate an understanding of the laws of logarithmic (c)
- use the chain rule to differentiate correctly (c)
- correctly square the function before integrating to find volume (d)
- include π in the solution (d)
- justify algebraic solutions to show they satisfy the given equation or state which cases are excluded (eii).

Areas for students to improve include:

- using factorisation to solve a trigonometric equation involving multiple trigonometric ratios (a)
- using the product rule or chain rule to differentiate a logarithmic function (c)
- using the derivative of logarithms in the previous section to integrate a logarithmic function (c)
- realising the connection between earlier parts of a question through the use of 'hence' (c)
- applying and using index laws (d)

- understanding that squaring a function will always give volumes greater than zero (d)
- having a range of methods to solve equations with absolute values (eii)
- including two valid reference points when graphing an absolute value function (ei)
- identifying endpoints with domain and range in the diagram (ei)
- sketching a diagram using at least $\frac{1}{3}$ of a page to show the scale, shape and significant points (ei).

Question 14

Students should:

- show the substitution of $t = 0$ into $v = \frac{1}{2}e^{2t} - 4t + c$ to evaluate c (a)
- factorise a non-monic trinomial to obtain solutions to equation (bi)
- show numerical results when testing stationary points using either the first or second derivative (bi)
- use a table with labelled rows, indicating x , $\frac{dy}{dx}$ and slope, when using the first derivative test (bi)
- integrate and substitute in (0,4) to calculate the constant (bii)
- provide information about the nature of the stationary points (biii)
- find and show the coordinates of stationary points (biii)
- calculate the interior or exterior angle of a polygon (c)
- calculate the equal angles in an isosceles triangle given the third angle (c)
- use exact ratios or the sine rule in trigonometry to evaluate the length of a side (c)
- obtain a derivative in terms of x with algebraic coefficients (d)
- obtain an equation in a and b using the derivative and the fact that the gradient of the tangent was 1 (d)
- obtain an equation in a and b using the function and the point of intersection of the tangent and curve (2, -2) (d)
- find expressions for gradient and y -coordinate, substitute into $y - y_1 = m(x - x_1)$ and equate coefficients with $y = x - 4$ (d).

In better responses, students were able to:

- integrate exponentials accurately (a)
- calculate the constant in an exponential equation (a)
- factorise a non-monic trinomial (bi)
- show values in their tests for nature of stationary points (bi)
- correctly calculate the constant (bii)
- calculate the y -coordinate at $x = -1$ in $y = x^3 + x^2 - x + 4$ (biii)
- match their information from (bi) to their graph (biii)
- understand that when $f''(x) = 0$ a curve is not concave down (biv)

- correctly name angles (c)
- substantiate all mathematical statements used (c)
- understand the need to prove $\angle DEX = 90^\circ$ (c)
- prove similar or congruent triangle with correct and relevant reasoning for each step in the proof (c)
- expand and equate coefficients (d).

Areas for students to improve include:

- understanding that velocity is the integral of acceleration (a)
- using $f'(x) = 0$ to find stationary points (bi)
- understanding the link between the sign of the second derivative and concavity (bi)
- understanding what a local maximum stationary point looks like graphically (biii)
- understanding that $f''(x) < 0$ for concave down (biv)
- solving linear inequalities accurately (biv)
- checking that all parts of a question are attempted (biv)
- setting out a proof in a logical order, stating reasons at each step (c)
- using simultaneous equations to achieve answers (d)
- understanding that two different pieces of information are needed to make two separate equations (d)
- understanding that changing the subject of an equation and then substituting that back into the same equation will not solve the required unknowns (d)
- understanding that y' finds gradients and y finds y -coordinates (d).

Question 15

Students should:

- use the laws of logarithms to change $2 \ln x$ to $\ln x^2$ (a)
- manipulate the equation to reach a quadratic equation (a)
- recognise the invalidity of a negative solution as $\ln x$ is only defined for $x > 0$ (a)
- use Pythagoras' theorem to find equations in p , q and h and solve for h (b)
- use matching sides of similar triangles to link p , q and h and solve for h (b)
- expand $(p + q)^2$ and make use of algebraic manipulation (b)
- use similar triangle proof showing clear steps (ci)
- extend BR and AQ to form a large triangle where AB is the hypotenuse to find D^2 (ci)
- differentiate using chain rule for D^2 (cii)
- test the nature of stationary point to show that it is a minimum (cii)
- explain why $x > 0$ for the context of this question (cii)
- use $P(\text{at least } 1) = 1 - P(\text{none})$ or the sum of the probabilities of each relevant outcome (di)
- use a probability tree to show outcomes (di)

- extrapolate the 'at least one concept' to n trials (dii)
- recognise that n must be an integer as it represents the number of people (dii).

In better responses, students were able to:

- understand that the domain of $\ln x$ is $x > 0$
- recognise the link between exponentials and logarithms (a)
- simplify $e^{2 \ln x}$ to get x^2 (a)
- clearly identify the triangle being worked with (ci)
- explain or show on diagrams what introduced variables are representing (ci)
- understand that it is possible to differentiate D^2 to obtain $\frac{dD^2}{dx}$ (cii)
- understand that it is valid to substitute $x = 2$ to show $\frac{dD^2}{dx} = 0$ (cii)
- solve an exponential inequation using logarithms (dii)
- recognise $\ln 0.98$ is negative and therefore reversing the inequality sign when dividing both sides of the equation by it (dii).

Areas for students to improve include:

- solving quadratic equations by factorisation (a)
- understanding the steps required to 'show' a result and showing clear working (b) and (cii)
- writing a correct similar triangle proof with clear reasoning (b)
- matching correct sides in similar triangles (b)
- showing working out steps (ci)
- using the mark allocation as a guide to the amount of work required (ci)
- understanding that minimums are obtained when the first derivative is equal to 0 (cii)
- carefully solving complex equations that involve algebraic fractions (cii)
- showing working for the testing of stationary points and making clear conclusions (cii)
- calculating the probability of a complementary event (di)
- understanding that showing the checking step is vital when using guess and check to solve equation (dii).

Question 16

Students should:

- find the amount remaining in the account after the withdrawal at the end of the fourth and eighth quarters by explicitly showing the progression from A_1 to A_2 (ai)
- present all working out with clear and sequential steps (ai)
- use patterns to generate an expression for A_n as a summation of terms (aii)
- use the correct value for the common ratio when finding the sum of a geometric progression formula (aii)
- quote the Reference Sheet for Simpson's rule and understand the meaning of $\frac{b-a}{6}$

compared to the alternative formula using $\frac{h}{3}$ (b)

- find the derivative of an expression with the treatment of the exponent of r (ci)
- realise that $1^{r+1} = 1$ to find the equation of the tangent (ci)
- use subtraction of areas to find the area of the given region (cii)
- use the quotient rule from the Reference Sheet, and remember to square the denominator (ciii)
- justify the maximum value by using the simplest method (ciii).

In better responses, students were able to:

- solve logarithmic equations (aii)
- understand the time when the particle comes to rest occurs when $v = 0$ (b)
- show substitutions into Simpson's rule (b)
- understand that the tangent meets the x -axis at $y = 0$ (ci)
- clearly present all the steps in a 'show' question (ci)
- solve quadratic equations (ciii)
- show numerical results when testing for concavity using the first derivative table (ciii).

Areas for students to improve include:

- understanding the steps required to 'show' a result (ai)
- improving accuracy and skill in algebraic manipulation of logarithmic equations (aii)
- understanding the difference between radians and degrees when using trigonometric functions (b)
- understanding that three function values are required for one application of the rule (b)
- simplifying expressions that involve a fractional common difference within a fraction (b)
- carefully examining the given diagram to select the most appropriate and simplest method (cii)
- considering the marks available for each part to gauge the required amount of working out (cii)
- using the answer provided in the question as a checking mechanism, rather than working backwards (cii)
- showing all algebraic manipulations in a 'show' question (cii).